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The British Tertiary Igneous Province: K–Ar Ages of Some Dykes and Lavas, from Mull, Scotland

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Summary

K–Ar age determinations of some of the oldest lavas, and of dykes from the regional swarm are considered together with geological, petrological and palaeomagnetic evidence. The apparent ages do not correlate with the polarity, and correlate only weakly with the degree of feldspar alteration. For the lavas alone there is a strong correlation with potassium content which is thought to be coincidental, the range of ages being due to argon loss.

The reversely magnetized plateau lavas were probably extruded within a period of less than 1 My, in the interval 60 ± 2 My ago. The dykes, of normal, reversed and shallow reversed polarities, probably were intruded during the interval 56–49 My ago.

1. Introduction

Samples have been selected for dating from collections of rocks whose palaeomagnetism has already been measured, with the object of determining their age within the British Tertiary volcanic sequence, and also to determine what time interval the palaeomagnetic results cover.

As all the available samples were altered at least slightly, a five-stage classification of alteration was established, with the immediate object of selecting the freshest samples, but also with the long-term aim of improving the selection criteria of samples for dating.

Samples were collected by drilling cores, as described in a paper by Ade-Hall *et al.* (1972) and only used for dating after their palaeomagnetism had been measured.

Argon was determined on a conventional extraction line, and potassium was measured by the method of standard addition using an EEL flame photometer. Further details may be found in a paper by Eurdy *et al.* (1972).

In the sections that follow, the petrographic classification and the geology of Mull are described preparatory to a discussion of the dating results.

2. Petrographic classification

The accepted selection criteria for dating whole rock samples is that they should be fresh and free of glass and zeolites. However, it is known that these requirements may be over-stringent (McDougall 1966), and as many cases exist—particularly

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in connection with palaeomagnetism—in which no completely fresh material is available it is important to refine the criteria. As a preliminary step we have quantified the alteration of the available Mull samples into five classes, with sub-divisions. As these rocks contain little obvious nepheline or glass, plagioclase feldspar was assumed to be the major potassium-bearing phase, and consequently the assessment was based primarily on this mineral. We consider that alteration, usually along cleavage directions, the growth of secondary minerals, e.g. zeolite, and the cracking of feldspars need to be evaluated, and accordingly every rock has been given a number under the following scheme: class 1: no alteration, minor cracking only allowed; class 2: minor alteration along cracks permitted; class 3: local replacement of feldspar in addition to 2. Usually major cracking; occasional zeolite-filled vesicles tolerated; class 4: major alteration; feldspar pseudomorphs still distinct, including some of the original material; discrete masses of zeolite common; and class 5: no recognizable feldspars remaining. Further, a letter was added to the class number to denote that the average state of the feldspars is fresher than that of the mafic minerals (a) or vice versa (b).

The freshest samples were selected for dating: these fell into classes 2 and 3, but in addition two samples of class 4 were added to see if their ages were significantly lower.

3. Evolution of the Mull volcano

The island of Mull consists almost entirely of a large, dissected volcano. The earliest volcanics exposed are the plateau lavas of the north-west and south-west, away from the central complex. These regions include the two areas of this study. In many places on Mull the Tertiary lavas lie on a thin but widespread mudstone, which is also found on the mainland in Ardnamurchan and the Loch Aline area, and in Skye, where it becomes tuffaceous. This mudstone is believed to derive from decomposition of volcanic ash, deposited shortly before the main lava eruptions (Bailey *et al.* 1924). At Ardtun, in the south-west, leaf-beds are intercalated near the base of the lavas. Gardner (1887) assigned them to the lower Eocene (the Palaeocene was not then recognized) and a Palaeocene age is now generally accepted, though Simpson (1961) placed them in the Miocene or lower Pliocene.

Evolution of the volcano led to the development of the central complex with its many intrusive features. The island is cut by many linear dykes, predominantly striking NNW–SSE, some of which extend far beyond Mull, north to the Outer Hebrides and south to Ayrshire, with a few extending even into Northern England. Cutting relations show that dyke formation must have continued during much of the development of the central complex. However, these cutting relations apply only to the central complex and there is no evidence to tell if the dykes away from the central complex—in particular, those of north-west Mull—were formed during a short interval or not. Richey (1961) considered that many of the dykes were formed towards the close of igneous activity. Walker (1970) deduced from the presence of the zeolite mineral laumontite that the areas of plateau lavas exposed at present had been hydrothermally metamorphosed to temperature of about 200 °C, presumably due to burial by about 2000 m of subsequent lavas, now eroded away. Ade-Hall *et al.* (1972) using Curie-point data, confirmed this heating in the dykes of the north-west, which cut the plateau lavas, and they suggested that the temperatures may have been a little higher towards the Ardmore Point end of the north-west peninsula. However, the dykes are generally less altered than the lavas and contain few zeolites, which suggests that the climax of thermal metamorphism had passed by the time the dykes considered here were intruded.

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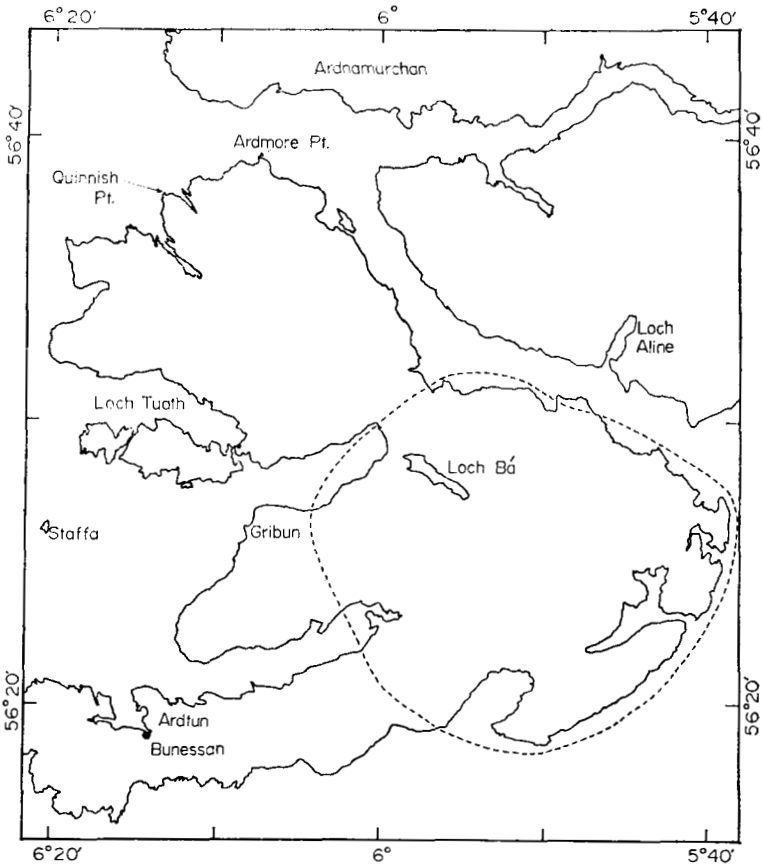


FIG. 1. Map of Mull showing localities referred to in text. The limit of pneumatolysis is shown by the dashed line.

4. Results and discussion

4.1. *Lavas*

The lavas, from Gribun, west Mull, are outside the zone of pneumatolysis associated with the central complex (Bailey *et al.* 1924, chapter VII) (see Fig. 1), and the lowest lava rests on mudstone.

The apparent ages (Table 1(a)) show a range of 46–63 My which does not represent the duration of extrusion because the ages are not in stratigraphic order. Instead, the duration may be estimated geologically. Although the lavas are all basalts with olivine and pyroxene, petrographic examination shows that the succession of 59 lavas (the numbering contains gaps) may be divided into six mineralogical groups. We propose, following the terminology of Nichols (1936), that each group is a lava flow comprising several flow units. For example, the bottom seven lavas (flow units) are all uniformly feldspar-rich alkali basalts, poor in olivine and relatively rich in potassium. Fractionation of olivine (low K_2O phase) in the magma chamber may be responsible for the variations between lava flows, and some indication of potassium variation can be seen in Table 1(a).

Flow units would be extruded in quick succession with no time for laterite formation: several years per group is a maximum time by comparison with modern volcanoes (Walker 1972). But vesicular lava top surfaces with associated weathering effects

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Table 1(a)

Gribun lavas

Sample	Alteration	Potassium, %K				$A^{40}_{rad}/g \times 10^6$ c.c.s.t.p.	$A^{40}_{rad}/A^{40}_{total}$ %	Age My
C 64–7a	3a	0·159	0·154	0·152	0·157	0·311	56·7	49·4±1·4
63–3a	2b	0·283	0·285	0·285		0·705	61·7	61·2±2·0
60–1a	4b	0·224	0·224	0·226	0·226	0·450	71·9	49·5±1·4
–3a	3b	0·277	0·271	0·260	0·270	0·517	64·7	47·3±1·3
						0·577	36·4	52·9±1·8
						0·523	66·0	48·0±1·5
58–1a	3a	0·246	0·244	0·249		0·630	74·2	63·2±2·3
54–2a	3b	0·259	0·256	0·250		0·632	61·2	61·2±2·0
26–3a	3a	0·204	0·204	0·198		0·394	54·0	48·3±1·6
6–3a	2a	0·988	1·015	1·025		2·36	83·5	57·8±2·0
5–5a	2a	1·085	1·075	1·060		2·49	70·9	57·4±2·0
4–1a	2a	1·070	1·055	1·030		2·47	74·2	58·2±2·3
–2a	3a	0·965	0·995	1·025		2·32	56·0	57·5±1·9
						2·49	36·6	61·7±2·0
3–1a	3a	1·097	1·082	1·098		2·70	72·8	60·6±3·0
–4a	3a	1·135	1·130	1·186	1·155	2·62	83·0	56·2±2·6
2–4a	3a	0·944	0·926	0·982	0·957	2·21	68·1	57·2±2·5
1–1a	3a	0·950	0·973	0·970		2·24	60·7	57·4±2·6

$\lambda = 5{\cdot}30 \times 10^{-11}\ \text{yr}^{-1}$ $\lambda_e = 0{\cdot}585 \times 10^{-11}\ \text{yr}^{-1}$ $K^{40}/K = 1{\cdot}19 \times 10^{-4}$ (atom ratio)

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Table 1(b)

North-west dykes

Sample	Alteration	Potassium, %K				$A^{40}_{rad}/g \times 10^6$ c.c.s.t.p.	$A^{40}_{rad}/A^{40}_{total}$ %	Age My
Normal dykes								
D 175–1a	2b	0·135	0·135	0·136		1·51	42·0	261·8± 9·6
–1b	2b	0·131	0·132	0·132		1·53	42·6	271·0±12·6
						1·45	41·8	256·7± 8·0
D 186–1a	4a	1·42	1·42	1·47	1·44	3·10	71·9	53·1± 2·4
						3·17	88·4	54·3± 1·8
D 264–3b	3a	1·055	1·018	1·022		2·05	58·7	49·2± 1·8
D 286–2b	3a	0·324	0·325	0·322	0·324	0·656	61·3	50·1± 1·4
Reversed dykes								
D 195–4b	3a	0·316	0·321	0·316		0·666	55·2	51·8± 1·5
D 213–1a	2a	0·128	0·129	0·131		0·277	61·5	53·1± 2·1
–3a	2a	0·116	0·112	0·114		0·252	46·1	54·6± 1·9
D 237–1a	2b	0·427	0·427	0·426		0·853	51·3	49·4± 1·4
D 259–2a	2a	0·137	0·142	0·141		0·317	44·1	56·0± 1·7
–3a	2a	0·130	0·128	0·132		0·286	50·2	54·5± 1·7
Shallow reversed dykes								
D 177–1a	3b	0·227	0·226	0·231	0·228	0·466	32·7	50·6± 1·5
D 235–1a	2b	0·186	0·191	0·193	0·190	0·394	31·7	51·3± 1·5
D 240–2a	2a	0·910	0·925	0·930		1·93	57·9	51·9± 1·7
D 284–2a	2a	0·128	0·128	0·126		0·283	46·6	55·2± 1·7

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and the development of laterite horizons, e.g. above unit 48, suggest there may have been considerable delay between successive lava flows (groups of flow units). Such time lapses most probably represent between 100 and 100 000 years, again by comparison with modern volcanoes.

Thus the total time for extrusion of all the lavas was not long and we consider that a million years is a reasonable upper limit (*cf.* Hawaii where 4200 m of lavas above sea-level have been formed in 0·8–1 My (McDougall 1964; Doell & Cox 1965).

The scatter of the apparent ages of the lavas will now be discussed in relation to four hypotheses. In Fig. 2 the apparent ages are plotted against potassium and fall into three distinct groups: high K, high age (group I), low K, high age (group II), and low K, low age (group III). The first hypothesis is that both low K groups have larger random error producing the total range of groups II and III, but this is belied both by the size of calculated errors and the tight grouping, for one would expect a continuous band of ages to result.

Systematic differences could account for the existence of the two low K groups but are not apparent. Mineralogically, the groups are similar; for instance, lavas 63 and 64 are very alike, yet have ages of 61 and 49 My. There may be a statistical difference of alteration between these two groups for the only class 4 sample is in group III and the only class 2 sample is in group II, but this can be connected to age only indirectly, for most samples of both groups are class 3. Palaeomagnetic directions and intensities were available but showed no systematic differences.

A third explanation involves excess argon in group II and argon loss in group III, relative to group I. Although excess argon has been found in lavas (McDougall *et al.* 1969; Dalrymple & Lanphere 1969) the amounts were very much smaller than needed to account for the differences of the means of groups II and III. However, this was checked by plotting a correlation diagram of $^{40}Ar/^{36}Ar$ *v.* $^{40}K/^{36}Ar$ (McDougall, Polach & Stipp 1969) which would be expected to give an intercept on the $^{40}Ar/^{36}Ar$ axis greater than the atmospheric ratio. As there are only three samples

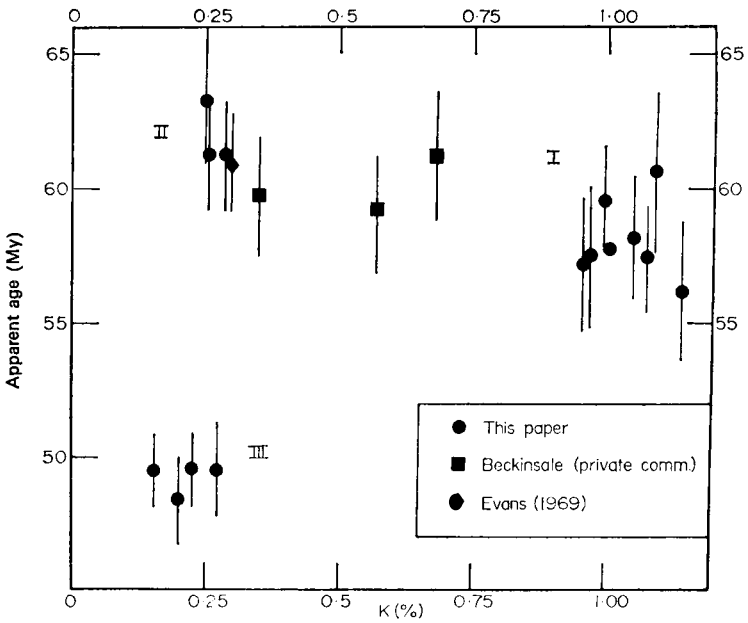


FIG. 2. Plot of K–Ar ages against potassium content, of Mull lavas.

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in group II a weighted best-line fit should not be taken too literally. It gave an intercept of 302·1 but the atmospheric ratio was well within the error, from which we conclude that though excess argon cannot be ruled out there is no evidence for it.

The final hypothesis is the familiar one of argon loss, with group II having lost least of the three groups, perhaps none. This is unsatisfactory in that one would expect a continuum of ages rather than the two distinct groups found. However, it is difficult to account for the low ages of group III in any other way and this explanation has the merit of employing only a single mechanism which is known to occur.

The Mull lavas have also been dated by other workers. Evans (1969) collected from a number of localities and his freshest sample, 'an olivine basalt with only a trace of sericite associated with the feldspars' came from Bunessan. This sample gave his oldest age: 60·9 ± 2 My, which he considered to be the best estimate of the age of crystallization. Beckinsale (private communication) sampled the lavas immediately above and below the Ardtun leaf beds, taking particular care to select fresh samples, and obtained dates respectively of 61·2 and 59·7 My. He also obtained an age of 59·2 My for a mugearite plateau lava near the head of Loch Tuath, about 8 km north of Gribun. Both his and Evans' data are shown in Fig. 2.

In the light of this discussion we conclude that most probably the range of apparent ages is due to argon loss and the distinctness of group III is fortuitous, though the other hypotheses cannot be entirely dismissed. The best estimate of the age of the lavas, bearing in mind the overlap of the error bars in Fig. 2, is 60 ± 2 My.

4.2. *North-west dykes*

The palaeomagnetic collection was made approximately between Ardmore Point and Quinnish Point, and measurements showed that these dykes could be divided into normal, reversed and shallow-reversed polarity groups. Details of sampling and results may be found in papers by Ade-Hall *et al.* (1969b), Ade-Hall & Wilson (1969), and Ade-Hall *et al.* (1972).

The K–Ar data for the dykes are set out in Table I(b): with the exception of D175; the apparent ages range between 49–56 My. There is no significant difference in age between the three palaeomagnetic groups. Petrological examination showed both analcite-bearing olivine dolerite and olivine tholeiite varieties of dykes, but there is no clear-cut distinction between the three polarity groups. As with the lavas, we must call upon other data to establish the plausibility or otherwise of these dates.

Evans (1969) has also dated the same dyke swarm and obtained his material from the same palaeomagnetic collection, but none of his dated dykes are in common with those of this paper. He found a range of 45–62 My and interpreted his values only as minimum ages. His figures cannot be taken at face value for several reasons. Firstly, two dykes were also dated by the ^{40}Ar–^{39}Ar step-heating method and gave higher ages, secondly, dyke D291 cuts D290, yet Evans obtained (57, 63) and 55·8 My respectively, the 63 My value being obtained by the ^{40}Ar–^{39}Ar method. Thirdly, Evans concluded that the most reliable age was that of the freshest sample, D271 which gave 62 My, which is approximately the same as the age of the plateau lavas. Though this age for the dykes cannot be ruled out definitely, it is unlikely, for within the central complex area many of the north–north-west trending dykes cut the Loch Bá felsite ring-dyke, itself known to be the youngest intrusion apart from the linear dykes (Bailey *et al.* 1924, chapter XXXIV). Evans concluded that the main period of dyke intrusion was between 65 and 55 My ago, probably in at least two pulses, and there may have been dyke intrusion as late as 44 My.

A different approach to estimating the duration of dyke formation derives from the palaeomagnetic investigation. Ade-Hall *et al.* (1972) found seven pairs of dykes that intersected, some of different polarities. They were able to show that the least number of periods of different polarity that would fit the pairs were two periods of

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reversed magnetization sandwiching a period of normal and a period of shallow reversed polarity—a total of four periods; however, they could not discount a larger number. Naidu (1971) investigated the average length of periods as a function of age and found that in the interval 56–48 My ago the periods were considerably longer than in more recent times, averaging 2 My. If this figure is applied to the four periods above, a figure of 8 My is found for the duration of dyke intrusion, which agrees with the range of ages found by us. But it must be emphasized that this is only a statistical estimate and the periods may have been much shorter than 2 My or, on the other hand, there may have been more than four periods of different polarity.

In the light of the above discussion we conclude tentatively that the dykes concerned were intruded in the interval 56–49 My ago, with the reservation that other dykes of the same swarm may have been intruded outside this interval.

Dyke D175 yielded an age of about 260 My on separate pieces of a single core. As these dykes cut the plateau lavas there is no question of the age being genuine; it is presumably due to excess argon. Excess argon is often due to high pressure during emplacement which retains magmatic gases including argon. Correspondingly the sample was examined in thin and polished sections for indications of retained gases in the form of large amounts of iron sulphide etc., but no abnormality was found. Alternatively, excess argon could be due to either the incorporation of an older xenolith, or to reheating at depth in a high partial pressure of argon.

The presence of large amounts of excess argon in one dyke raises the doubt of small amounts in others, and might account for the high ages of some of Evans's dykes.

5. Conclusions

The lava succession of Gribun, formed at an early stage in the evolution of the Mull volcano, was probably extruded in a period of not more than a million years, about 62 My ago.

The north-west dyke swarm seems to have been formed in the interval 56–49 My ago, but this conclusion is tentative. It has not been possible to assign any particular dyke to normal, reversed or shallow reversed polarity groups by its age or petrology.

The very great length of the dykes in the NNW and SSE directions suggests that the dyke swarm is not intimately connected with the Mull volcano, but rather the dykes and the volcanic centres which lie along this line—Mull, Skye, Arran etc.—are the expression of a regional tension, possibly connected with the opening of the North Atlantic. Recently Duncan, Peterson & Hargraves (1972) have put forward the suggestion that the volcanic centres of Antrim and western Scotland are due to relative motion of a tectonic plate and a plume or 'hot spot'. This hypothesis was applied specifically to the volcanic centres and casts no light on the origin of the markedly linear dykes associated with the centres.

Finally, a subsidiary aim of this work was to examine the petrological criteria for selection of rocks for K–Ar dating. It is evident from the data of Table I(a) that the ages of the samples of class 2 are closely grouped near to the maximum observed, whereas those of class 3 give markedly variable ages. The data of this paper are too few to make any more firm conclusions and the class scheme outlined earlier will be refined in the light of further experience.

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References

Ade-Hall, J. M., Dagley, P., Wilson, R. L., Evans, A. L. & Riding, A., 1969a. Palaeomagnetism of the Mull (Scotland) Lavas. *Trans. Am. geophys. Un.*, **50,** 132.

Ade-Hall, J. M., Dagley, P., Wilson, R. L., Smith, P. J., Evans, A. & Riding, A., 1969b. A palaeomagnetic study of the Mull dyke swarm, *Palaeogeophysics*, ed. Runcorn S.K., Academic Press, London.

Ade-Hall, J. M., Dagley, P., Wilson, R. L., Evans, A., Riding, A., Smith, P. J., Skelhorne, R. & Sloan, T., 1972. A palaeomagnetic study of the Mull regional dyke swarm, *Geophys. J. R. astr. Soc.*, **27,** 517.

Ade-Hall, J. M. & Wilson, R. L., 1969. Opaque petrology and natural remanence polarity in Mull (Scotland) dykes. *Geophys. J. R. astr. Soc.*, **18,** 333.

Bailey, E. B., Clough, T. C., Wright, W. B., Richey, J. E. & Wilson, G. V., 1924. Tertiary and post-Tertiary geology of Mull, Loch Aline and Oban, *Mem. geol. Surv., Scotland*, H.M.S.O.

Dalrymple, G. B. & Lanphere, M. A., 1969. *Potassium-argon dating*, Freeman, San Francisco.

Doell, R. R. & Cox, A., 1965. Palaeomagnetism of Hawaiian lava flows, *J. geophys. Res.*, **70,** 3377.

Duncan, R. A., Peterson, N. & Hargraves, R. B., 1972. Mantle plumes, movement of the European plate and polar wandering, *Nature*, **239,** 82.

Evans, A. L., 1969. *On dating the British Tertiary igneous province*, Ph.D. Thesis, Cambridge University.

Gardner, J. S., 1887. On the leaf-beds and gravels of Ardtun, Carsaig etc., in Mull, *Q. Jl geol. Soc. Lond.*, **43,** 270.

McDougall, I., 1964. K–Ar ages of lavas of the Hawaiian Islands, *Bull. geol. Soc.Am.*, **75,** 107.

McDougall, I., 1966. Precision methods of potassium-argon isotopic age determination on young rocks, *Methods and techniques in geophysics.*, vol. 2, ed. S. K. Runcorn, Interscience, London.

McDougall, I., Polach, H. A. & Stipp, J. J., 1969. Excess radiogenic argon in young subaerial basalts from the Auckland volcanic field, New Zealand, *Geochim. Cosmochim. Acta*, **33,** 1485.

Naidu, P. S., 1971. Statistical structure of geomagnetic field reversals, *J. geophys. Res.*, **76,** 2649.

Nichols, R. L., 1936. Flow units in basalt, *J. Geol.*, **44,** 617.

Purdy, J. W., Mussett, A. E., Charlton, S. R., Eckford, M. & English, H., 1972. The British Tertiary igneous province: potassium-argon ages of the Antrim basalts, *Geophys. J. R. astr. Soc.*, **27,** 327.

Richey, J. E., 1961. Scotland: the Tertiary volcanic districts, *British Regional Geology*, H.M.S.O. Edinburgh.

Simpson, J. B., 1961. The Tertiary pollen-flora of Mull and Ardnamurchan, *Trans. R. Soc. Edin.*, **64,** 421.

Walker, G. P. L., 1970. The distribution of amygdale minerals in Mull and Morven (Western Scotland), *West Commemoration Volume*, page 181.

Walker, G. P. L., 1972. Compound and simple lava flows and flood basalts, *Bull. Volcanol.*, in press.

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